作业 5

1 .

（1）

$$q(x) = K\frac{h_1^2 - h_2^2}{2L} - \frac{1}{2}WL + Wx$$

$$q(0) = 14.5 \times \frac{17.6^2 - 15.3^2}{2 \times 525} - \frac{1}{2} \times 0.007 \times 525 = -0.79\ m^2/d$$

（2）

$$q(525) = 14.5 \times \frac{17.6^2 - 15.3^2}{2 \times 525} - \frac{1}{2} \times 0.007 \times 525 + 0.007 \times 525 = 2.88\ m^2/d$$

（3）

$$a = \frac{L}{2} - \frac{K}{W}\frac{h_1^2 - h_2^2}{2L} = 113.2\ (m)$$

（4）

$$h_{\max}^2 = h_1^2 + \frac{h_2^2 - h_1^2}{L}a + \frac{W}{K}(La - a^2)$$

$$h_{\max} = 17.8\ (m)$$

2．

（2）

$$a = \frac{L}{2} - \frac{K}{W}\frac{h_1^2 - h_2^2}{2L} = 1000\ (m)$$

得

$h_2^2 - h_1^2 = 200\ (m)$，$h_2 = 24.495\ (m)$

3．

$$a = \frac{L}{2} - \frac{K}{W}\frac{h_1^2 - h_2^2}{2L} = -478.07\ (m)$$

$$q(L) = K\frac{h_1^2 - h_2^2}{2L} - \frac{1}{2}WL + WL = 7.5 \times 10^{-6}$$

$$Q = q(L) \times B = 2.1 \times 10^{-2}\ m^3/s$$

4．

$$\Delta(h_{0,t}^2) = 28^2 - 15^2 = 559$$

$$h_{A,t}^2 - h_{A,0}^2 = 25^2 - 15^2 = 400$$

$$F(\lambda) = 0.7156$$

反查表得

$$\lambda = 0.258$$

$$t = \frac{x_A^2}{4a\lambda_A^2} = 2347\ d$$

可以编程计算

```
from scipy.optimize import root

from scipy import special

def func(x):

return special.erfc(x)-0.7156

sol=root(func, 0.5 )

sol.x
```

$$\lambda = 0.2576$$

$$t = \frac{x_A^2}{4a\lambda_A^2} = 2354\ d$$